

E-Commerce EDA Assignment

Mohammad Afzal Malik
Class Roll No: 24
University Roll No: 2315510125
GLA University

Phase 1: Data Audit

The dataset contains 500,000 records and 30 columns. Some columns had missing values such as shipping cost, rating and delivery days. These missing values were handled before further analysis.

Phase 2: Pricing Data Skewness

I used a histogram and boxplot to analyze the final price column. Most products have lower prices while only a few products have very high prices. The skewness value was 2.11, showing that the data is right-skewed.

Phase 3: Categorical Columns

The dataset contains categorical columns like category, city, payment method and loyalty tier. These columns cannot be directly used in machine learning models, so encoding was required.

Phase 4: Return Analysis

Out of 500,000 orders, around 59,904 orders were returned. This gives a return rate of nearly 12%. Different categories showed similar return patterns.

Phase 5: Feature Engineering

Missing values were filled using median values. The return_reason column was filled with 'No Return'. Categorical columns were encoded and a new feature called price_per_item was created.

Phase 6: Correlation Analysis

A heatmap was created to understand the relationship between numerical features. Price and delivery-related features may have an impact on product returns.

Conclusion

The dataset was cleaned and analyzed successfully. Product prices were highly skewed, and missing values and categorical data were handled properly. The dataset is now ready for machine learning and further analysis.